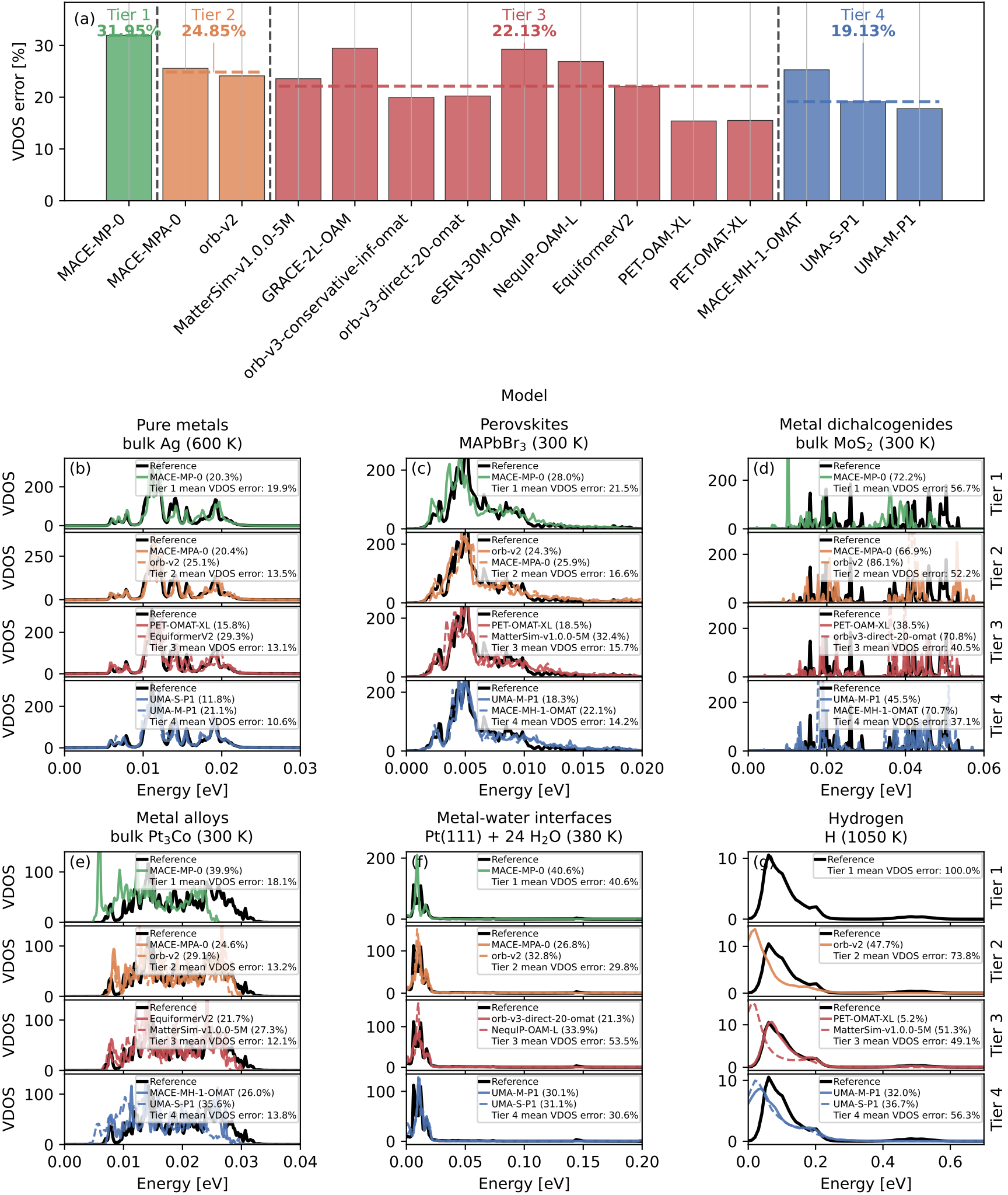

(a)
Tier 1
31.95%
Tier 2
24.85%
Tier 3
22.13%
Tier 4
19.13%
VDOS error [%]
MACE-MP-0
MACE-MPA-0
orb-v2
MatterSim-v1.0.0-5M
GRACE-2L-OAM
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1
UMA-M-P1
Model
Pure metals
bulk Ag (600 K)
Perovskites
MAPbBr3 (300 K)
Metal dichalcogenides
bulk MoS2 (300 K)
(b)
Reference
MACE-MP-0 (20.3%)
Tier 1 mean VDOS error: 19.9%
MACE-MPA-0 (20.4%)
orb-v2 (25.1%)
Tier 2 mean VDOS error: 13.5%
PET-OMAT-XL (15.8%)
EquiformerV2 (29.3%)
Tier 3 mean VDOS error: 13.1%
UMA-S-P1 (11.8%)
UMA-M-P1 (21.1%)
Tier 4 mean VDOS error: 10.6%
(c)
MACE-MP-0 (28.0%)
Tier 1 mean VDOS error: 21.5%
orb-v2 (24.3%)
MACE-MPA-0 (25.9%)
Tier 2 mean VDOS error: 16.6%
PET-OMAT-XL (18.5%)
MatterSim-v1.0.0-5M (32.4%)
Tier 3 mean VDOS error: 15.7%
UMA-M-P1 (18.3%)
MACE-MH-1-OMAT (22.1%)
Tier 4 mean VDOS error: 14.2%
(d)
MACE-MP-0 (72.2%)
Tier 1 mean VDOS error: 56.7%
MACE-MPA-0 (66.9%)
orb-v2 (86.1%)
Tier 2 mean VDOS error: 52.2%
PET-OAM-XL (38.5%)
orb-v3-direct-20-omat (70.8%)
Tier 3 mean VDOS error: 40.5%
UMA-M-P1 (45.5%)
MACE-MH-1-OMAT (70.7%)
Tier 4 mean VDOS error: 37.1%
VDOS
Energy [eV]
Tier 1
Tier 2
Tier 3
Tier 4
Metal alloys
bulk Pt3Co (300 K)
Metal-water interfaces
Pt(111) + 24 H2O (380 K)
Hydrogen
H (1050 K)
(e)
MACE-MP-0 (39.9%)
Tier 1 mean VDOS error: 18.1%
MACE-MPA-0 (24.6%)
orb-v2 (29.1%)
Tier 2 mean VDOS error: 13.2%
EquiformerV2 (21.7%)
MatterSim-v1.0.0-5M (27.3%)
Tier 3 mean VDOS error: 12.1%
MACE-MH-1-OMAT (26.0%)
UMA-S-P1 (35.6%)
Tier 4 mean VDOS error: 13.8%
(f)
MACE-MP-0 (40.6%)
Tier 1 mean VDOS error: 40.6%
MACE-MPA-0 (26.8%)
orb-v2 (32.8%)
Tier 2 mean VDOS error: 29.8%
orb-v3-direct-20-omat (21.3%)
NequIP-OAM-L (33.9%)
Tier 3 mean VDOS error: 53.5%
UMA-M-P1 (30.1%)
UMA-S-P1 (31.1%)
Tier 4 mean VDOS error: 30.6%
(g)
Tier 1 mean VDOS error: 100.0%
orb-v2 (47.7%)
Tier 2 mean VDOS error: 73.8%
PET-OMAT-XL (5.2%)
MatterSim-v1.0.0-5M (51.3%)
Tier 3 mean VDOS error: 49.1%
UMA-M-P1 (32.0%)
UMA-S-P1 (36.7%)
Tier 4 mean VDOS error: 56.3%